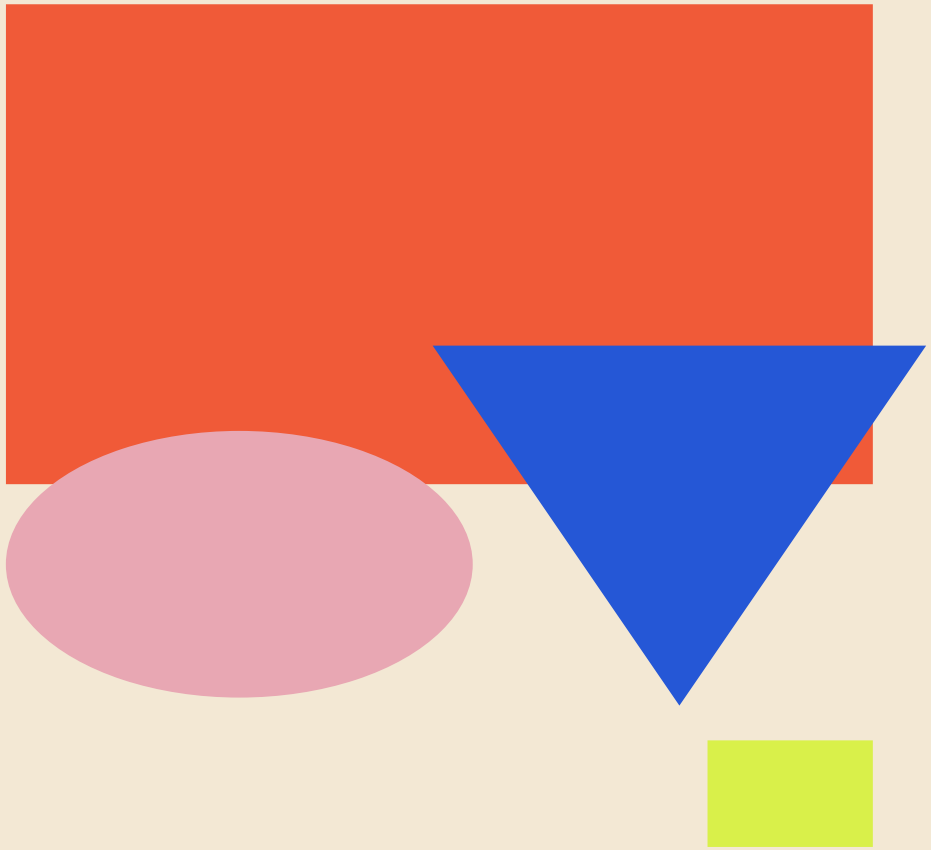
DATA SYSTEMS ADMINISTRATOR · ENTRY-LEVEL COURSE

WINDOWS FUNDAMENTALS



Files, folders, drives, paths —

and your first commands.

4 TOPICS · 1 HANDS-ON LAB · 0 PREREQUISITES

COURSE MAP

FOUR STOPS. ONE NEW SKILL.

01 FILE TYPES

Read any file name and know what it will do.

exit skill · spot .exe, .zip, .iso and say what each is for

02 FILES & FOLDERS

Create, name, move, and inspect anything.

exit skill · manage files by rule, not by guesswork

03 DRIVES & PATHS

Navigate disks, USB sticks, and addresses.

exit skill · find any file fast and eject media safely

04 COMMAND LINE

Type your first CMD and PowerShell commands.

exit skill · do file jobs with dir, cd, copy — and a cmdlet

TOPIC 01

01

UNDERSTANDING FILE TYPES

What .exe, .zip, and .iso actually mean —
and why Windows reads the label before you do.

The extension is a label Windows actually reads

report — the name you choose

.docx — the type Windows reads

- Windows uses the extension to pick the **icon** you see and the **app that opens** the file when you double-click it.
- Extensions are **hidden by default**. Turn them on: File Explorer → **View** → **Show** → **File name extensions**.
- Renaming *holiday.zip* to *holiday.exe* does **not** convert it. The inside of the file never changes.

TRY IT Turn extensions on, then open your Downloads folder and look again.

Source: Microsoft Support — Common file name extensions in Windows (2025)

THE USUAL SUSPECTS

Five families you will meet every day.

PROGRAMS

.exe

.msi .bat

They **run** things — installers, apps, and command scripts.

CONTAINERS

.zip .rar .iso .cab

They pack things — open or extract first.

DOCUMENTS

.txt .csv .pdf

.docx .xlsx .pptx

You read and edit them — school and office staples.

MEDIA

.jpg .png .mp3 .mp4

You watch and listen — photos, music, videos.

SYSTEM

.dll .sys

Windows' own parts — **leave them alone.**

Source: Microsoft Support — Common file name extensions in Windows (2025)

.zip packs it small. .iso clones the whole disc.

.zip

THINK: A SUITCASE

Squeezes files smaller so they are easier to store and send.

Extract first — right-click → Extract All before you actually use what is inside.

Windows opens .zip by itself; .rar needs an extra tool.

.iso

THINK: A PHOTOCOPY OF A WHOLE DISC

An exact image of a CD or DVD, packed into one file.

Double-click to mount — Windows pretends it is a real disc drive with its own letter.

This is how Windows install media is usually shared.

Also in the family: .cab — Microsoft's own "cabinet" archives, used for Windows updates and drivers. You will see them, but rarely open them yourself.

Source: Microsoft Support — Common file name extensions in Windows (2025)

Double-click means "run this."

- **.exe** and **.msi** are programs and installers — when they open, they **do** things to the computer.
- **.bat** and **.cmd** are scripts: plain text files full of commands that run one after another.

THE CLASSIC TRICK

photo.jpg.exe

With extensions hidden, Windows shows only **photo.jpg** — the dangerous .exe stays invisible.

Attackers rename malware this way and email it as a "picture" or "invoice".

QUICK CHECK A file shows up as invoice.pdf.exe — what is its real type?

THE RULE

Only run executables from sources you **trust**.

Unexpected attachment? Weird download? **Stop and ask** before double-clicking.

Source: Microsoft Learn — File Types (Win32); Pooj & Patil, file-upload security study (2016)

TOPIC 02

02

FILES & FOLDERS IN ACTION

Create it. Name it. Move it.

And read its ID card — Properties.

Windows has house rules for names

NEVER IN A FILE NAME

\ / : * ? " < > |

- **Reserved names** are off-limits too: CON, PRN, AUX, NUL, COM1–COM9, LPT1–LPT9. Windows owns these words.
- **Length limit:** the full path — every folder plus the file name — should stay under **260 characters**.
- **Capital letters don't matter:** Report.docx and REPORT.docx are the same file to Windows.

SAFE HABIT Stick to letters, numbers, spaces, hyphens (-) and underscores (_) — always legal.

Source: Microsoft Learn — Naming Files, Paths, and Namespaces (2023)

Create, move, rename, delete — and why a file sometimes refuses

THE FIVE MOVES

New folder · right-click → New, or Ctrl+Shift+N

Copy · Ctrl+C, then Ctrl+V — original stays

Move · Ctrl+X, then Ctrl+V — or drag it

Rename · select it, press F2

Delete · Delete key → Recycle Bin (undoable)

Shift+Delete · skips the Bin — gone for good

WHEN A FILE FIGHTS BACK

It's open in an app. Locked files can't move or delete — close the app first.

The path is too long. Shorten folder names, or move it closer to the drive root.

You lack permission. Some folders need an admin — that's normal, not a bug.

Technician's order: close it → shorten it → check permissions.

Source: Microsoft Learn — Can't delete files on an NTFS file system

Properties is the file's ID card

Right-click any file → Properties. Two tabs matter most.

GENERAL TAB · FACTS & FLAGS

The facts: file type, location, size, and when it was created and last modified.

The flags (attributes):

Read-only — look, don't edit

Hidden — invisible unless you choose to show hidden items

The **attrib** command shows the same flags in CMD.

SECURITY TAB · WHO MAY DO WHAT

NTFS permissions decide **who** can do **what**:

Read → open and view

Write → change the contents

Modify → change and delete

Full control → everything, including permissions

The **icacls** command shows the same list in CMD.

Don't mix them up: attributes (read-only, hidden) describe the **file**. Permissions describe what **people** may do with it.

Sources: Microsoft Learn — attrib; Microsoft Learn — icacls (2023)

TOPIC 03

03

DRIVES, MEDIA & PATHS

Where things live — disks, USB sticks, and the addresses that find them.

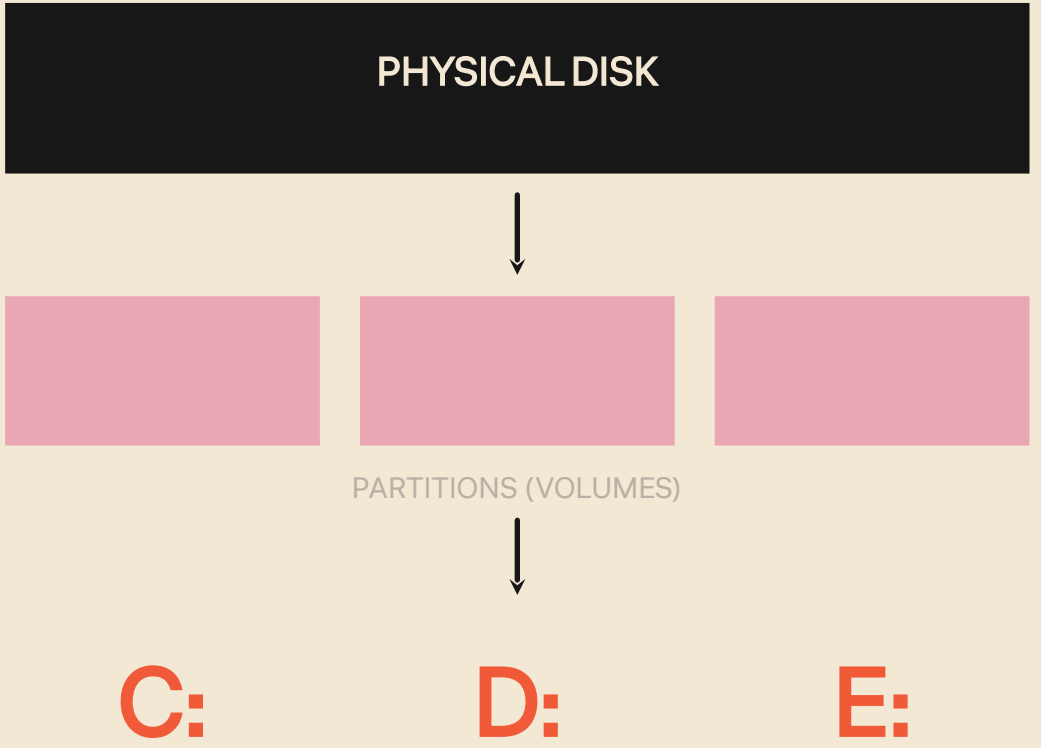
One disk can wear many letters

A **physical disk** is the hardware inside (or plugged into) the computer.

It gets split into **partitions** — the usable sections Windows calls **volumes**.

Each volume gets a **drive letter**: C:, D:, E: …

C: is where Windows and your installed programs live, by default.



Plug in a USB stick — it simply takes the next free letter.

The tool: Disk Management (diskmgmt.msc) — view disks, assign letters, initialize new drives.

Source: Microsoft Learn — Overview of Disk Management (2024)

NTFS, FAT32, exFAT — same shelf, different rules

The file system is the set of rules a drive uses to store files. You choose one when formatting.

FILE SYSTEM	NTFS	FAT32	exFAT
USED FOR	The Windows system drive (C:)	Old devices, small sticks	Modern USB sticks & SD cards
BIGGEST FILE	Huge — no practical limit	**4 GB max** — a real limit	Huge — no practical limit
SUPERPOWER	Permissions + journaling (self-repair)	Works on almost everything	Big files + works everywhere

REMEMBER Formatting a USB stick for class or work? Pick exFAT — big files, works on Windows and Mac.

Source: Microsoft Learn — NTFS overview (2023)

TOPIC 03 · DRIVES & PATHS

Every file has an address — learn to read it

C:\Users\Ava\Documents\report.docx

drive letter · folders, separated by \ · the file itself

ABSOLUTE PATH

The full address, starting at the drive letter.

C:\Users\Ava\Documents\report.docx

Works from anywhere.

RELATIVE PATH

Directions from where you are standing.

Documents\report.docx

Only works if you're already in C:\Users\Ava.

UNC PATH · NETWORK

An address on another computer.

\\SERVER\share\file.txt

Two backslashes = "find this machine on the network".

REMEMBER Same 260-character limit applies to the whole path — deep folder nests cause problems.

Source: Microsoft Learn — Naming Files, Paths, and Namespaces (2023)

Eject first. Your files will thank you.

WHY EJECTING MATTERS

Windows doesn't always write to a USB stick **immediately** — it holds data in a cache to work faster.

Pull the stick out mid-write, and the file can end up **half-written and corrupted**.

THE SAFE ROUTINE

1 · Click **Safely Remove Hardware** in the taskbar corner

2 · Choose your device, wait for the "safe to remove" message

3 · Only then pull it out

"DEVICE IS IN USE"?

Something still has a file open on the stick.

Close its files and apps, then eject again. Don't just yank it.

TECHNICIAN'S WARNING

Never plug in a USB drive you found.

Unknown sticks are a classic malware delivery trick — attackers leave infected drives where people will pick them up.

Found one at work or school? Hand it to IT. Don't "check what's on it".

Sources: Microsoft Support — Safely remove hardware in Windows; Pham et al., USB malware study (2010)

TOPIC 04

04

THE COMMAND LINE

CMD and PowerShell —
same jobs, no clicking.

CMD: a few words do the work

Open it: press Win + R, type cmd, hit Enter.

```
C:\Users\Ava> dir

C:\Users\Ava> cd Documents

C:\Users\Ava\Documents> md Project

C:\Users\Ava\Documents> copy notes.txt Project\

C:\Users\Ava\Documents> ren notes.txt old-notes.txt

C:\Users\Ava\Documents> dir /?
```

Type a command, press Enter, read what it says back.

STARTER VOCABULARY

dir — list what's here

cd — change folder (cd .. goes up)

md — make a folder · **rd** — remove it

copy / **move** — duplicate / relocate files

del — delete · **ren** — rename

cls — clear the screen

command /? — built-in help for anything

REMEMBER dir and /? will never hurt anything.

Source: Microsoft Learn — Windows Commands (A–Z) (2024)

PowerShell: same idea, superpowers

CMD SAYS

```
dir
copy notes.txt Project\
```



POWERSHELL SAYS

```
Get-ChildItem
Copy-Item notes.txt Project\
```

Cmdlets read like sentences: **Verb-Noun**. Learn the pattern once, guess commands forever.

WHY ADMINS LOVE IT

CMD passes text between commands.

PowerShell passes objects — real, structured data — so commands can chain into pipelines that automate whole jobs.

Scripts: .bat for CMD, .ps1 for PowerShell.

Get-Help explains any cmdlet — the manual lives inside.

REMEMBER Not either/or: both shells are already on every Windows PC — use the right one for the job.

Source: Microsoft Learn — What is PowerShell? (2025)

One job, three ways

The job: create a folder called Invoices and move Q1.xlsx into it.

WITH THE MOUSE

1 · Right-click empty space → New → Folder

2 · Type Invoices, press Enter

3 · Drag Q1.xlsx onto the folder

IN CMD

```
md Invoices

move Q1.xlsx Invoices\
```

Two lines. Done in five seconds.

IN POWERSHELL

```
New-Item -ItemType Directory
Invoices

Move-Item Q1.xlsx Invoices\
```

Same result, every time. For one file, the mouse is fine. For a hundred files, or the same job every Monday — typing wins.

Sources: Microsoft Learn — Windows Commands; PowerShell Management cmdlets

PRACTICE SCENARIO

YOUR TURN: THE 15-MINUTE LAB

Work inside your own practice folder — nothing in this lab can break Windows.

1	Turn on file name extensions, then open Downloads and write down the type of five files you find there.	**Done when:** you can say what each of the five files is for
2	With the mouse, build Projects\2026\Jan, move any file into Jan, and rename it — following the naming rules.	**Done when:** the path reads Projects\2026\Jan\yourfile, no banned characters
3	Do task 2 again — no mouse. Use CMD (md, cd, move), then list the result in PowerShell with Get-ChildItem.	**Done when:** both shells show the same folder structure

Stuck? That's the job — use command /? or Get-Help, then ask. Technicians read manuals.

TAKE IT WITH YOU

POCKET REFERENCE

NAMES

Banned: \ / : * ? " < > |

Reserved: CON, PRN, AUX, NUL, COM1–9, LPT1–9

Limit: 260 characters for the full path

Case: doesn't matter

Renaming ≠ converting — the inside never changes

PLACES

C: — Windows and programs

USB stick — takes the next free letter

exFAT — right choice for removable media

\\SERVER\share — network address

Eject first — always

Found a USB drive? — hand it to IT, never plug it in

COMMANDS

dir · cd · md · rd

copy · move · del · ren

command /? — help in CMD

Get-ChildItem · New-Item · Copy-Item · Move-Item

Get-Help — help in PowerShell

WRAP-UP

YOU CAN NOW...

- Read any file name and say what it will do
- Create, name, move, and inspect files and folders
- Find anything on any drive — and eject USB safely
- Run your first CMD and PowerShell commands

KEEP LEARNING

Microsoft Learn — free official docs and tutorials

Professor Messer — free CompTIA A+ video course

CompTIA A+ Core 2 — the exam these topics map to

NEXT CLASS

Hands-on lab: user folders, permissions, and your first script.

PRACTICE · EXPLORE · ASK QUESTIONS